

Dongxu Shen

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EDUCATION

Hong Kong University of Science and Technology (Guangzhou)

Sep 2023 – Present

B.S. in Artificial Intelligence

GPA: 3.3 / 4.3

Relevant Coursework: Data Structures and Algorithms (A), Machine Learning (A-),
Introduction to AI (A), Mathematics for AI (A)

University of California, Berkeley

Sep 2025 – Dec 2025

Exchange Student

GPA: 3.9 / 4.0

Relevant Coursework: Computer Vision(A), Natural Language Processing(A)

RESEARCH INTERESTS

My research interests lie in 3D vision and neural rendering, with a long-term goal of building scalable and photorealistic 3D representations for real-world scene reconstruction.

RESEARCH

Phoneme-Aware Audio-to-Mesh Facial Animation

Apr 2026 – Present

Research Intern under Dr. Xiaoli Liu, developing phoneme-aware models for speech-driven 3D facial animation.

- Aiming to develop semantically meaningful viseme-aware 3DMM mouth bases and phoneme-conditioned audio-to-mesh models to improve speech-specific facial articulation beyond PCA-derived expression spaces.

Unposed 3D Gaussian Head Avatar Reconstruction

Jan 2026 – Apr 2026

Research intern under Dr. Xiaoli Liu, focusing on reconstructing high-quality and animatable 3D Gaussian head avatars from unposed multi-view data.

- Identified inaccurate camera calibration in existing 3D head avatar datasets such as EmoTalk3D, we proposed the first calibration-free framework for reconstructing animatable 3D Gaussian head avatars from unposed multi-view data.
- Designed a robust FLAME pose initialization strategy, enabling FLAME training from unposed multi-view data.
- Implemented joint optimization of Gaussians, FLAME parameters, and camera poses, and designed a tri-plane-based view-consistent Gaussian color representation, surpassing existing baselines on novel view synthesis by over 5 dB PSNR.

Large-Scale 3DGS SLAM

Sep 2025 – Feb 2026

Advised by Prof. Hao Wang at HKUST(GZ). Worked on monocular 3D Gaussian Splatting SLAM for large-scale outdoor scenes.

- Proposed KiloGS-SLAM, a monocular 3DGS-SLAM framework for kilometer-scale outdoor scenes, addressing pose tracking failure and memory overhead in long-sequence reconstruction.
- Designed an image-texture-complexity-aware Gaussian initialization strategy, and developed multi-view-consistency-guided Gaussian densification and pruning methods for robust large-scale scene reconstruction.

Whole Slide Image Classification

May 2025 – Aug 2025

Research intern with Prof. Shidang Xu at South China University of Technology, focusing on whole slide image classification.

- Addressed the domain gap introduced by small-patch WSI representations by designing a large-patch feature extractor.
- Led baseline reproduction, experimental evaluation, ablation studies, and paper figure visualization to support the empirical validation of the proposed framework.

PUBLICATIONS & MANUSCRIPTS

- **AnyAvatar: High-Fidelity Gaussian Head Avatars under Uncalibrated Camera Settings**
ACM International Conference on Multimedia (ACM MM) 2026 (Under Review)
 Yujian Liu*, **Dongxu Shen***, Haoran Li*, Yuting Liu, Chuang Chen, Xinyi Jiang, Zhupeng Jiang, Peng Cao, Shidang Xu, Xiaoli Liu[†].
- **Robust and Efficient Monocular 3D Gaussian SLAM for Kilometer-Scale Outdoor Scenes**
European Conference on Computer Vision (ECCV) 2026 (Under Review)
 Sicheng Yu*, **Dongxu Shen***, Beizheng Zhao, Guanzhi Ding, Hao Wang[†].
- **Minimal High-Resolution Patches Are Sufficient for Whole Slide Image Representation via Cascaded Dual-Scale Reconstruction**
The Chinese Conference on Pattern Recognition and Computer Vision (Under Review) [\[arXiv\]](#) [\[Project Page\]](#)
 Yujian Liu*, Yuechuan Lin*, **Dongxu Shen***, Haoran Li, Yutong Wang, Xiaoli Liu, Shidang Xu[†].
- **MoGaFace: Momentum-Guided and Texture-Aware Gaussian Avatars for Consistent Facial Geometry**
The Chinese Conference on Pattern Recognition and Computer Vision (Under Review) [\[arXiv\]](#) [\[Project Page\]](#)
 Yujian Liu*, **Dongxu Shen***, Chuang Chen, Zairan Wang, Linlang Cao, Fanyu Geng, Peng Cao, Shidang Xu, Xiaoli Liu.

SKILLS

Programming Languages	Python, C++, JavaScript (TypeScript), Bash/Shell scripting, CUDA
Mathematics	Calculus, Linear Algebra, Complex Functions, Convex Optimization.
Platforms & Tools	Windows, Linux; VSCode, Git, L ^A T _E X, Markdown, Blender, MeshLab
Research Areas	3D Gaussian Splatting, 3D Reconstruction, Neural Rendering, 3D Head Avatars, Visual SLAM, Speech-driven Facial Animation